

remain at the same position as shown.

- 8 (a) (i)** Conservation of linear momentum  
 $0 = -(A - 4)v + 4V$

**(ii)** From (i), hence

$$\begin{aligned}
\frac{KE_{\alpha}}{KE_{daughter}} &= \frac{\frac{1}{2}(4u)V^2}{\frac{1}{2}(A-4)uV^2} \\
&= \frac{4V^2}{(A-4)V^2} \\
&= \frac{4\left(\frac{1}{4}(A-4)v\right)^2}{(A-4)v^2} \\
&= \frac{1}{4}(A-4) \\
&= \frac{1}{4}A - 1
\end{aligned}$$

**(b) (i)** Change in mass  
 $= (211.9459u) - (207.9374 + 4.0015)u$   
 $= 0.007u$   
 $= 1.162 \times 10^{-29} \text{ kg}$   
 $E = \Delta mc^2$   
 $= (1.162 \times 10^{-29})(3.00 \times 10^8)^2$   
 $= 1.0458 \times 10^{-12} \text{ J}$   
 $= 6.54 \text{ MeV}$

**(ii)** To find energy alpha particle

$$\begin{aligned}
\frac{KE_{\alpha}}{KE_{daughter}} &= \left(\frac{1}{4}A - 1\right) \\
&= \left(\frac{1}{4}(212) - 1\right) \\
&= 52 \\
\text{Hence } KE_{\alpha} &= \left(\frac{52}{53}\right)(6.54 \text{ MeV}) = 6.42 \text{ MeV}
\end{aligned}$$

**(c) (i)** The energy released is not purely in the form of kinetic energy of the daughter nucleus and alpha particle. Some energy is released as gamma radiation.

**(ii)** The gamma photon has a small but non-zero momentum according to the de broglie's wavelength.

The gamma photon move off in a direction that is not collinear with that of the thalium nucleus and alpha particle, carrying with it a component of momentum perpendicular to the velocities indicated.

Since momentum must be conserved in the perpendicular direction as well, the thalium nucleus and alpha particle cannot move off in opposite direction.

**(d) (i)**

| nucleus | Half – life /s | Decay constant/s <sup>-1</sup> |
|---------|----------------|--------------------------------|
|---------|----------------|--------------------------------|

|              |      |                      |
|--------------|------|----------------------|
| Bismuth-212  | 3600 | $1.9 \times 10^{-4}$ |
| Thallium-208 | 190  | $3.7 \times 10^{-3}$ |

(ii) This is because Thallium-208 is not stable but will decay to form another nucleus lead-208.

Since the half-life of thallium-208 is much shorter than that of bismuth-212, the rate of conversion of thallium into lead is much greater than the rate of conversion of bismuth into thallium. This accounts for the very small amount of thallium present after 2 hours.

Note: Most answers involve large difference between two half-lives but do not assume alpha particles are part of count for number of nuclei.